

PAYSTONE EXPLAINABLE AI

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# Bank Customer Churn Analysis

*Understanding, Predicting, and Preventing Customer Attrition*

**Prepared for Executive Leadership**

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## Executive Summary

PayStone serves 10,127 credit-card customers, of whom 1,627 (16.1%) have churned. This analysis examines who churns, the behavioural factors that distinguish churners from retained customers, and an AI model designed to identify at-risk customers before they leave, creating an opportunity for proactive intervention.

|                                  |                                |                            |                                               |
|----------------------------------|--------------------------------|----------------------------|-----------------------------------------------|
| <b>10,127</b><br>Total Customers | <b>1,627</b><br>Customers Lost | <b>16.1%</b><br>Churn Rate | <b>92.5%</b><br>Best Model Accuracy (ROC-AUC) |
|----------------------------------|--------------------------------|----------------------------|-----------------------------------------------|

## Key Findings

- **Engagement, not demographics, drives churn:** The strongest predictors of attrition are how often a customer contacts support, how inactive they have become, and how many products they hold, not age, income, or education.
- **Churners disengage before they leave:** Customers who eventually churn hold fewer products on average (**3.28 vs. 3.91**), show declining transaction activity, and have significantly lower card utilisation (**16% vs. 30%**).
- **A few segments carry elevated risk:** Platinum cardholders (**25.0%**) and Doctorate-level customers (**21.1%**) have churn rates well above PayStone's **16.1% overall rate**, despite representing relatively small segments.
- **The predictive model performs strongly:** CatBoost achieved a **92.5% ROC-AUC**, demonstrating strong ability to distinguish between customers likely to churn and those likely to remain.
- **The card portfolio is highly concentrated:** **93.2%** of customers hold the entry-level Blue card, making this the largest segment and the primary opportunity for broad-reaching retention initiatives.

## Recommended Next Steps

- **Deploy the churn model for monthly risk scoring:** Score the active customer base regularly and route high-risk customers into targeted retention programmes.

- **Prioritise early re-engagement:** Focus outreach on customers showing warning signs such as increased support contact frequency, declining transaction activity, and reduced product usage.
- **Investigate the Platinum and Doctorate-education segments:** Although these are relatively small segments, their elevated churn rates warrant further investigation through targeted customer feedback, exit surveys, and qualitative analysis.
- **Use product cross-selling as a retention lever:** Encourage customers to adopt additional relevant products, as higher **Total Relationship Count** is consistently associated with lower churn risk.



## 1. Business Context & Objective

**Customer attrition:** when a customer closes their account or stops using their card, this signifies one of the most direct threats to recurring revenue in the cards business. Acquiring a new customer is significantly more expensive than retaining an existing one, which makes early identification of at-risk customers a high-value capability.

**This project had two goals:** first, to understand the underlying drivers of churn using PayStone's historical customer data; and second, to build a predictive model capable of flagging customers who are likely to churn, so that the retention team can intervene proactively rather than reactively.

The analysis is based on a dataset of 10,127 customers, combining demographic information (age, gender, education, marital status, income), product holdings (card type, number of products), and behavioural data (transaction activity, credit utilisation, service contact history).

## 2. Who Are PayStone's Customers?

Before looking at churn specifically, it is useful to understand the overall shape of the customer base. The charts below summarise PayStone's 10,127 customers across the key demographic and product dimensions.

### 2.1 Overall Churn Snapshot

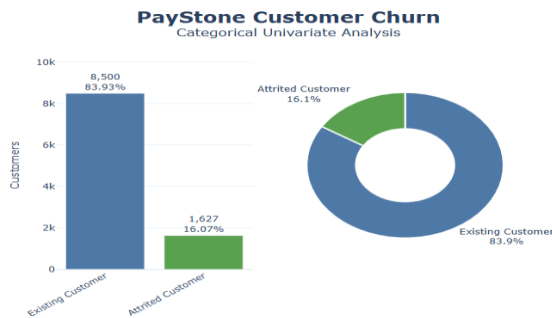


Figure 1. Of PayStone's 10,127 customers, 8,500 (83.9%) remain active while 1,627 (16.1%) have churned.

### 2.2 Gender & Education

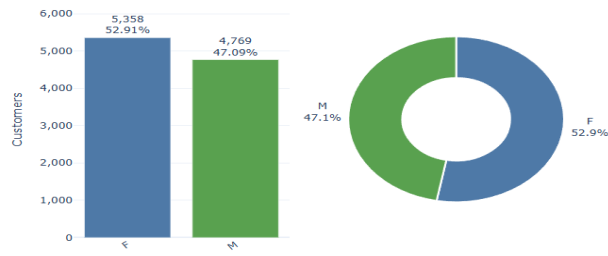


Figure 2. The customer base is evenly split by gender: 52.9% female, 47.1% male.

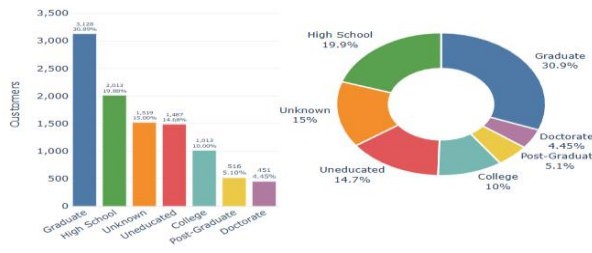


Figure 3. Nearly a third of customers (30.9%) hold a Graduate-level education, followed by High School (19.9%).

## 2.3 Marital Status & Income

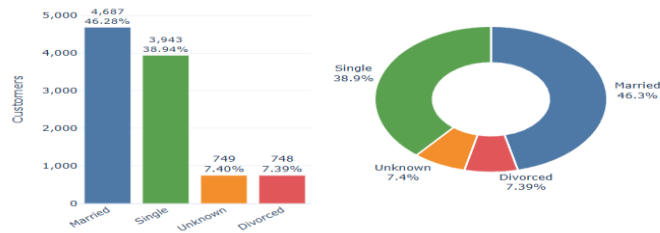


Figure 4. Married customers make up the largest group (46.3%), followed by Single customers (38.9%).

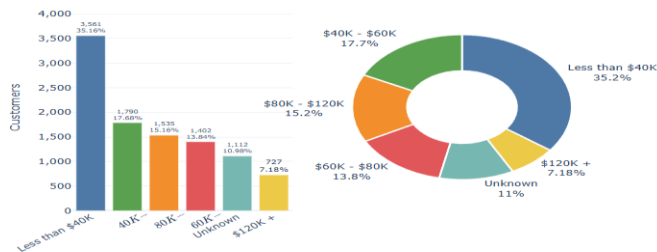


Figure 5. Over a third of customers (35.2%) report household income under \$40K, PayStone's largest income segment.

## 2.4 Product Mix

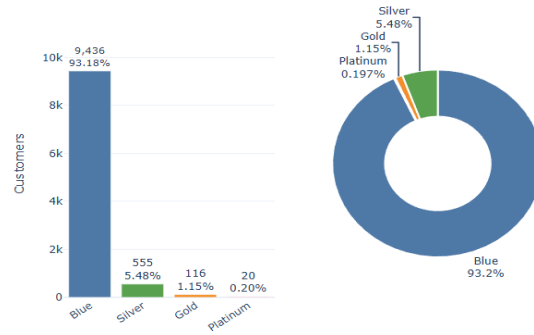


Figure 6. The Blue card dominates the portfolio at 93.2% of customers, with Silver, Gold, and Platinum together making up under 7%.

### 3. What Distinguishes Customers Who Churn?

To identify customers most at risk, we compared churn rates across key customer segments. The analysis shows that demographic differences in churn exist but are relatively modest, while behavioural and engagement factors provide much stronger signals of churn risk. These behavioural drivers are explored in greater detail in Section 4.

#### 3.1 Churn Rate by Demographic Segment

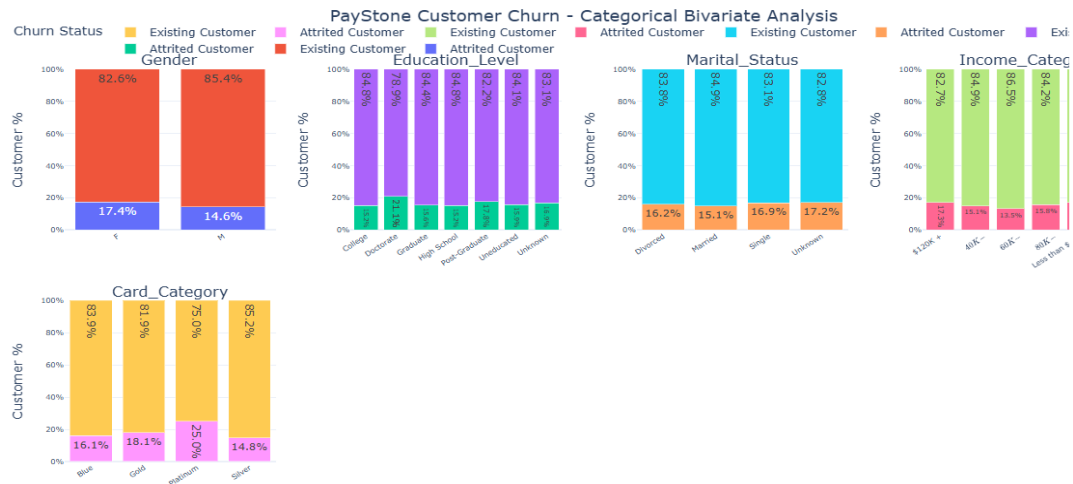


Figure 7. Churn rate by gender, education, marital status, income, and card type. Percentages shown are the share of each group that churned.

| Segment   | Group    | Churn Rate | vs. PayStone Base Rate (16.1%) |
|-----------|----------|------------|--------------------------------|
| Card Type | Platinum | 25.0%      | +8.9 pts (highest risk)        |

| Segment        | Group         | Churn Rate | vs. PayStone Base Rate (16.1%) |
|----------------|---------------|------------|--------------------------------|
| Card Type      | Gold          | 18.1%      | +2.0 pts                       |
| Education      | Doctorate     | 21.1%      | +5.0 pts                       |
| Education      | Post-Graduate | 17.8%      | +1.7 pts                       |
| Income         | \$120K+       | 17.3%      | +1.2 pts                       |
| Marital Status | Unknown       | 17.2%      | +1.1 pts                       |
| Gender         | Female        | 17.4%      | +1.3 pts                       |
| Card Type      | Silver        | 14.8%      | -1.3 pts (lowest risk)         |
| Income         | \$60K–\$80K   | 13.5%      | -2.6 pts                       |

#### Reading this table

Demographic differences in churn are present, but the gaps are generally modest, with most segments only a few percentage points above or below the 16.1% base rate. Demographics alone are therefore not strong predictors of churn. Platinum and Doctorate segments are notable exceptions, showing materially higher churn rates, although their smaller sample sizes warrant further validation before major strategic decisions are made.

## 4. The Real Warning Signs: Customer Behaviour

While demographics provide useful context, customer behaviour and engagement are much stronger indicators of churn. How customers use their cards, engage with PayStone, and interact with its services provides a clearer distinction between customers who stay and those who leave. The chart below compares average behavioural metrics for churned versus existing customers to highlight the patterns associated with customer attrition.

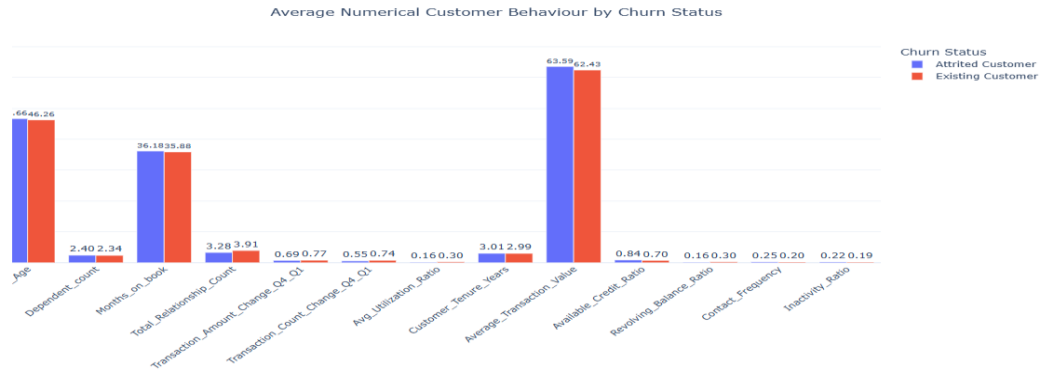


Figure 8. Average behavioural metrics: Attrited (blue) vs. Existing (red) customers.

#### 4.1 The Clearest Signals

| Behaviour                                | Churned Customers | Active Customers | What it tells us                                      |
|------------------------------------------|-------------------|------------------|-------------------------------------------------------|
| Products held (Total Relationship Count) | 3.28              | 3.91             | Churners use fewer PayStone products                  |
| Card utilisation rate                    | 16%               | 30%              | Churners use their available credit far less          |
| Revolving balance ratio                  | 16%               | 30%              | Churners carry a much smaller balance                 |
| Transaction count change (Q4 vs Q1)      | 0.55              | 0.74             | Churners' transaction activity drops off more sharply |
| Transaction amount change (Q4 vs Q1)     | 0.69              | 0.77             | Churners' spend also declines more                    |
| Support contact frequency                | 0.25              | 0.20             | Churners contact customer service more often          |
| Inactivity ratio                         | 0.22              | 0.19             | Churners have more inactive months                    |

Together, these findings reveal a clear behavioural pattern: customers who churn tend to disengage before leaving. They hold fewer products, utilise less of their available credit, and show declining transaction activity, while simultaneously contacting customer support more frequently, potentially indicating dissatisfaction or emerging service issues. In contrast, age, tenure, and average transaction

value show relatively limited differences between churned and existing customers. Overall, the evidence indicates that customer engagement and behavioural changes are stronger indicators of churn risk than demographic characteristics.

## 5. Building a Churn Prediction Model

To move from understanding historical churn to predicting future customer attrition, we developed and compared four machine learning models: CatBoost, XGBoost, LightGBM, and Random Forest. Each model was trained using the same customer data and evaluated on a held-out test set of previously unseen customers, ensuring a fair and consistent comparison of predictive performance. The models learn patterns that distinguish churned from retained customers, including the behavioural and engagement indicators identified in Section 4, and use these patterns to estimate each customer’s churn probability on a scale from 0 to 1. A higher probability indicates a greater likelihood of future churn.

### 5.1 Model Comparison

We evaluated each model based on its ability to discriminate between customers at higher and lower risk of churn, using ROC-AUC. This standard classification metric ranges from 50% (no better than random guessing) to 100% (perfect discrimination), with higher scores indicating stronger predictive performance.

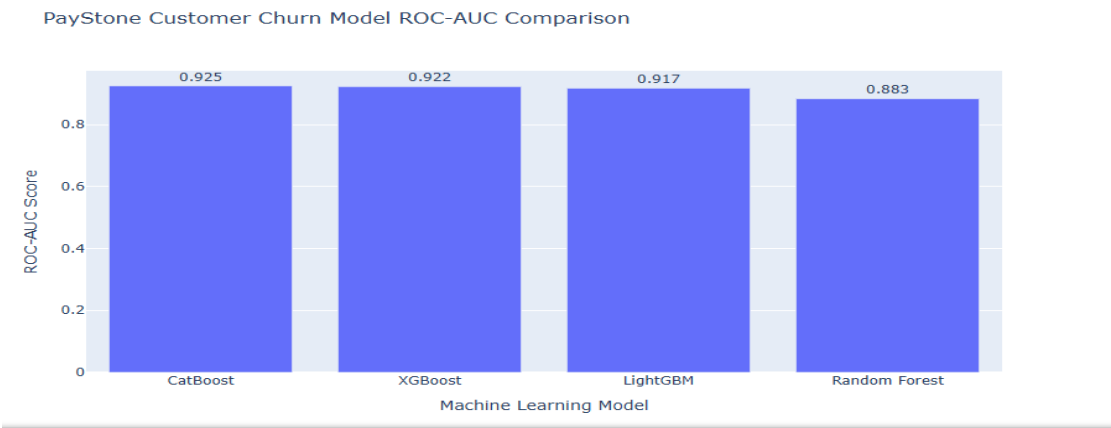


Figure 9. Model comparison by ROC-AUC score. All four models comfortably outperform random guessing (50%).

| Model    | ROC-AUC Score | Overall Accuracy | Verdict                       |
|----------|---------------|------------------|-------------------------------|
| CatBoost | 92.5%         | 91.2%            | Best performing — recommended |
| XGBoost  | 92.2%         | 90.9%            | Strong alternative            |

| Model         | ROC-AUC Score | Overall Accuracy | Verdict                        |
|---------------|---------------|------------------|--------------------------------|
| LightGBM      | 91.7%         | 90.4%            | Solid, slightly behind leaders |
| Random Forest | 88.3%         | 88.0%            | Weakest of the four            |

CatBoost achieved the strongest overall performance, delivering the highest ROC-AUC and the best balance between precision and recall, as reflected by the highest weighted F1-score (see Appendix). XGBoost performed comparably and represents a strong alternative model. Based on its superior predictive performance and overall robustness, CatBoost is recommended as the production model, with XGBoost retained as a validated fallback.

## 6. How Well Does the Model Perform in Practice?

A ROC-AUC score tells us how well a model ranks risk in general. To understand what that means for day-to-day use, we look at how the recommended model (CatBoost) performs on actual customers in our test set of 2,026 customers.

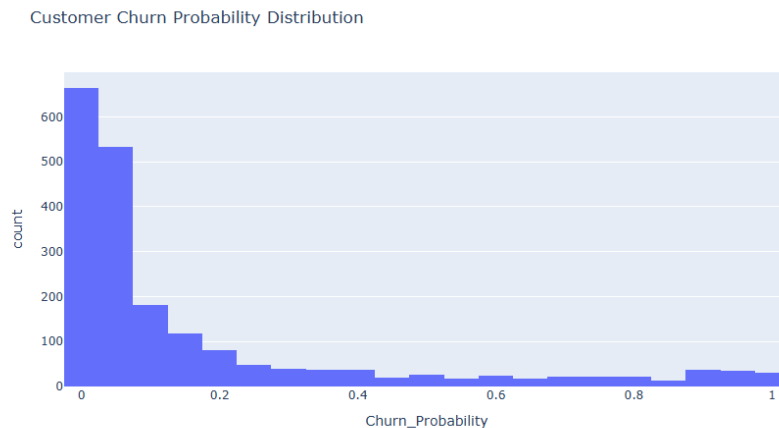


Figure 10. Distribution of predicted churn-risk scores. Most customers cluster at very low risk, with a smaller group of clearly high-risk customers — the model separates the two groups well rather than leaving everyone in an ambiguous middle range.

### 6.1 Catching Churners: The Confusion Matrix

Among the 325 customers who actually churned in the test set, CatBoost correctly identified 204, achieving a 62.8% recall (catch rate). It also correctly classified 1,653 of 1,701 retained customers as low risk, corresponding to a 97.2% true-negative rate. Together, these results demonstrate the model's ability to identify a substantial proportion of future churners while maintaining strong discrimination among customers who remain active.



Figure 11. Predicted vs. actual outcomes for all four models. CatBoost (top left) correctly identifies the most churners (204) of any model tested.

### What this means for the business

No predictive model will identify every future churner, and some false negatives are expected in any real-world classification system. The key finding is that CatBoost identifies approximately 6 in 10 customers who subsequently churn, providing PayStone with a meaningful early-warning capability across a test population of more than 2,000 customers. By integrating these risk scores into the retention workflow, PayStone can prioritise a targeted, high-risk customer segment for intervention rather than applying retention activity across the entire customer base. This enables a more focused and scalable approach to customer retention.

## 7. Key Drivers of Customer Churn

Beyond predicting which customers are most likely to churn, the model also provides insight into the factors driving those predictions. This explainability analysis reinforces and further refines the behavioural patterns identified in Section 4, helping translate model outputs into actionable customer retention insights.

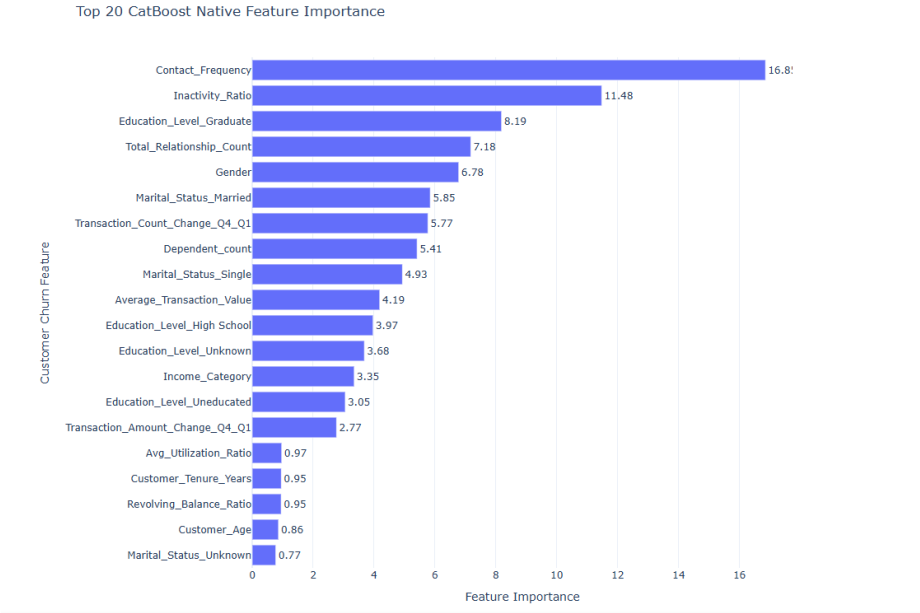


Figure 12. The factors the model relies on most heavily when predicting churn, ranked by importance.

Customer contact frequency and inactivity are the strongest predictive signals, carrying substantially greater influence than other factors in the model. This reinforces a clear and actionable insight: customers who become increasingly inactive or contact support more frequently should be prioritised for monitoring and early retention intervention.

7.1 Top Churn Drivers, Ranked

| Rank | Driver                     | What it means                                                                        |
|------|----------------------------|--------------------------------------------------------------------------------------|
| 1    | Contact Frequency          | How often a customer contacts customer service — the single strongest signal of risk |
| 2    | Inactivity Ratio           | How much of the past year the customer was inactive                                  |
| 3    | Education Level (Graduate) | Whether the customer holds a Graduate-level qualification                            |
| 4    | Total Relationship Count   | How many PayStone products the customer holds                                        |
| 5    | Transaction Count Change   | How much transaction activity has changed recently                                   |

8. Recommendations & Next Steps

## 8.1 Operationalise the Model

- Score the active customer base monthly using the CatBoost model and prioritise the highest-risk customers for targeted retention interventions.
- Set a risk-score threshold, such as the top 15–20% of customers, to balance retention team capacity with the potential cost of missing high-risk churners.

## 8.2 Act on the Early Warning Signs

- Proactively engage customers with rising support contact frequency, triggering a check-in before concerns escalate into formal complaints or account closure.
- Monitor transaction activity quarter-over-quarter, treating sustained declines in transaction count or spending as early warning indicators requiring further investigation.
- Use product cross-selling as a retention lever by encouraging relevant additional products for single-product customers, as broader product relationships are associated with stronger customer retention.

## 8.3 Investigate High-Risk Segments

- Conduct exit interviews or surveys with recently churned Platinum and Doctorate-education customers to understand whether their needs and expectations are being met by the current product offering.
- Review the Platinum card's value proposition to assess whether its benefits, pricing, and features align with the expectations and needs of the customers it attracts.

## 8.4 Suggested Timeline

| Phase | Action                                                     | Owner             | Timeframe  |
|-------|------------------------------------------------------------|-------------------|------------|
| 1     | Deploy monthly scoring pipeline                            | Data Science      | 0–1 month  |
| 2     | Build retention outreach workflow for high-risk customers  | CRM / Retention   | 1–2 months |
| 3     | Launch qualitative review of Platinum & Doctorate segments | Customer Insights | 1–2 months |
| 4     | Review model performance & retrain on new data             | Data Science      | Quarterly  |

## Appendix: Supporting Technical Detail

This appendix contains additional detail for technical stakeholders who wish to review the full model diagnostics.

### A.1 ROC-AUC Curves

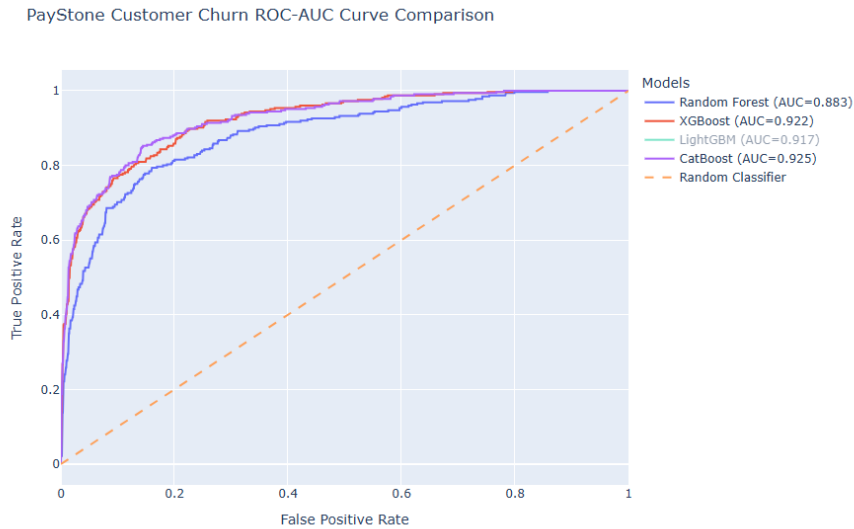


Figure A1. ROC curves for all four models. Curves closer to the top-left corner indicate better performance.

### A.2 Classification Reports

| PayStone Customer Churn Classification Reports |           |        |          |         |
|------------------------------------------------|-----------|--------|----------|---------|
| CatBoost                                       |           |        |          |         |
| Class                                          | Precision | Recall | F1-Score | Support |
| Macro Avg                                      | 0.871     | 0.800  | 0.829    | 2026    |
| Weighted Avg                                   | 0.912     | 0.917  | 0.912    | 2026    |

| XGBoost      |           |        |          |         |
|--------------|-----------|--------|----------|---------|
| Class        | Precision | Recall | F1-Score | Support |
| Macro Avg    | 0.866     | 0.793  | 0.823    | 2026    |
| Weighted Avg | 0.909     | 0.914  | 0.909    | 2026    |

| LightGBM     |           |        |          |         |
|--------------|-----------|--------|----------|---------|
| Class        | Precision | Recall | F1-Score | Support |
| Macro Avg    | 0.854     | 0.788  | 0.816    | 2026    |
| Weighted Avg | 0.904     | 0.910  | 0.905    | 2026    |

| Random Forest |           |        |          |         |
|---------------|-----------|--------|----------|---------|
| Class         | Precision | Recall | F1-Score | Support |
| Macro Avg     | 0.788     | 0.767  | 0.777    | 2026    |
| Weighted Avg  | 0.880     | 0.884  | 0.882    | 2026    |

Figure A2. Precision, recall, and F1-score for each model (macro and weighted averages).

### Key Findings

- CatBoost is the strongest overall model, achieving the highest macro precision (0.871), recall (0.800), and F1-score (0.829).
- XGBoost is a strong fallback, performing closely behind CatBoost with a macro F1-score of 0.823.

- LightGBM performs reasonably well, achieving a macro F1-score of 0.816, but trails the top two models.
- Random Forest is the weakest performer, with a macro F1-score of 0.777 and only 132 of 325 churners identified.
- Macro metrics are more informative for PayStone because they give equal importance to churned and existing customers, avoiding the distortion caused by class imbalance.
- CatBoost provides the best business value, identifying more potential churners while maintaining a strong balance between precision and recall.
- Recommendation: Deploy CatBoost as the production model and retain XGBoost as a validated fallback.

### A.3 SHAP Feature Importance

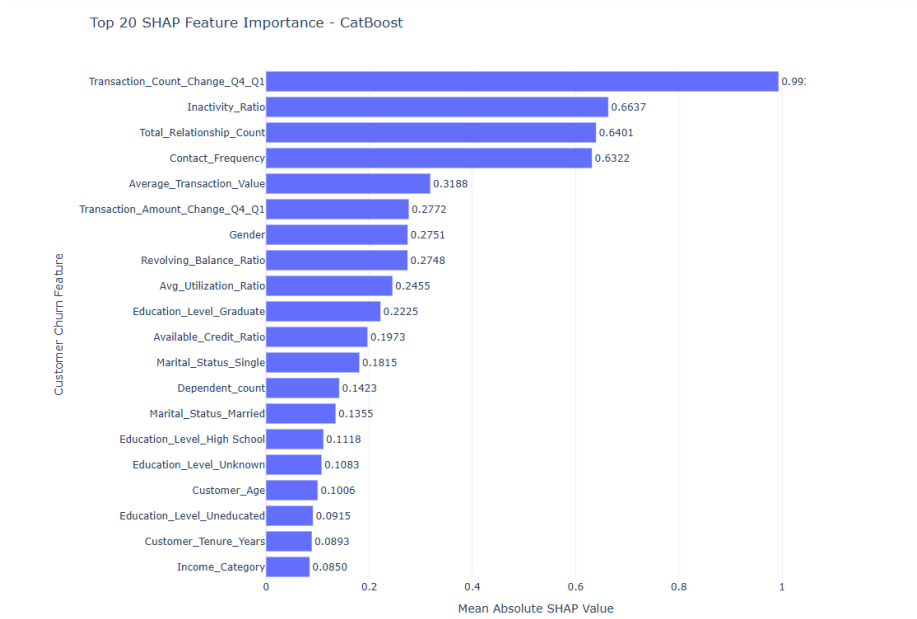


Figure A3. SHAP-based feature importance for CatBoost, an alternative importance measure that confirms behavioural features (transaction change, inactivity, relationship count, contact frequency) dominate the model.

- Transaction count change between Quarter1 and Quarter 4 is the dominant churn driver: With a mean absolute SHAP value of 0.993, it is the single most influential feature in the model.
- Changes in transaction activity are the strongest signal: The change in transaction count between Quarter1 and Quarter 4 indicates that declining customer activity is a key signal of churn risk.
- Inactivity ratio, total relationship count, and contact frequency form the second tier: Their mean absolute SHAP values are 0.664, 0.640, and 0.632, respectively.

- Behavioural variables dominate the SHAP rankings: Transaction activity change, inactivity, number of products held, and customer contact frequency are the four strongest predictors.
- Other behavioural and financial variables have lower influence: Average transaction value, gender, available credit ratio, and revolving balance ratio contribute less to the model's predictions.
- Demographic variables have comparatively limited influence: Education level and income category rank near the bottom of the top 20 features.
- Overall finding: Both SHAP and native feature importance consistently show that customer behaviour and engagement are substantially stronger predictors of churn than demographic characteristics.

## A.4 Feature Correlation Matrix

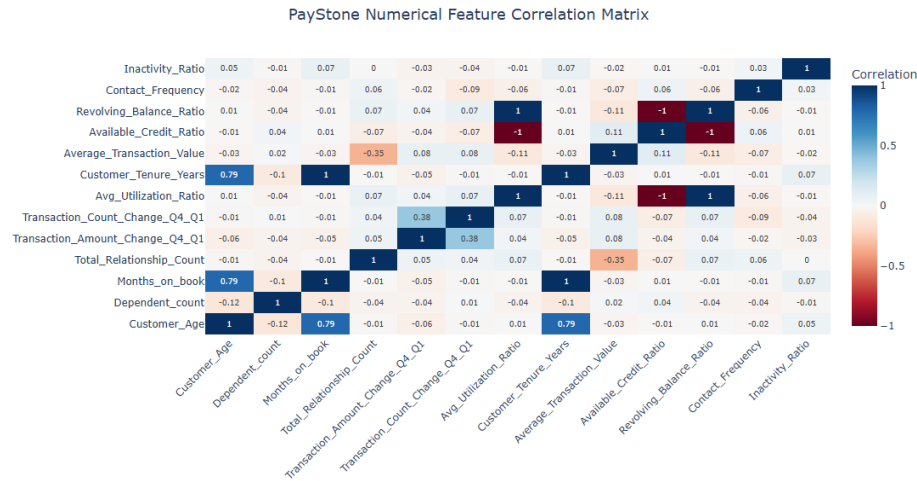


Figure A4. Correlation matrix of numerical features, used to check for redundancy between engineered variables.

- Customer age, months on book, and customer tenure are strongly correlated at approximately 0.79, which is expected because customers with longer relationships with PayStone tend to be older.
- Average credit utilisation and revolving balance ratio are perfectly correlated (1.00), indicating that they capture essentially the same underlying information.
- Average credit utilisation and available credit ratio have a perfect negative correlation (-1.00), reflecting their mathematical relationship: as credit utilisation increases, the proportion of available credit decreases.
- Average credit utilisation, revolving balance ratio, and available credit ratio therefore contain substantial redundancy. Retaining all three provides limited additional information and could be reduced in a future model iteration.

- Most other numerical features show weak correlations, generally below  $\pm 0.10$ , suggesting that they capture largely distinct aspects of customer behaviour.
- Overall, the feature set shows limited multicollinearity, with the main redundancy concentrated among the three-credit utilisation and availability measures.

## A.5 Demographic Segmentation by Churn Status



Figure A5. Raw customer counts split by churn status across each demographic dimension.

### Customer Segment Distribution and Business Implications

- Graduate is the largest education segment, with approximately 3,100 customers. For PayStone, this means retention strategies targeting this group could reach a substantial portion of the customer base, although its churn rate should be considered alongside its large population size.
- Doctorate customers show a higher proportion of churn, despite representing a much smaller segment. Their 21.1% churn rate suggests a potential retention opportunity that warrants further investigation rather than broad strategic changes based on segment size alone.
- Married and Single customers represent the largest marital-status groups, giving PayStone the greatest potential reach when designing retention initiatives around these segments.
- The “Less than \$40K” income category is substantially larger than the other income groups, making it an important segment for understanding customer behaviour and designing appropriately targeted products and engagement strategies.

- The Blue card category dominates the product portfolio, with most customers holding blue cards. This means that retention initiatives focused on blue cardholders would have the broadest reach and potentially the greatest overall business impact.
- Segment size and churn rate should be considered together: A small segment with a high churn rate may represent a targeted opportunity, while a large segment with a moderate churn rate can represent a much larger absolute number of customers at risk.
- For PayStone, the key implication is prioritisation: Use customer volume to determine where retention initiatives can have the greatest reach, and churn rates to identify where targeted interventions may be most valuable.

## A.6 Numerical Feature Distributions

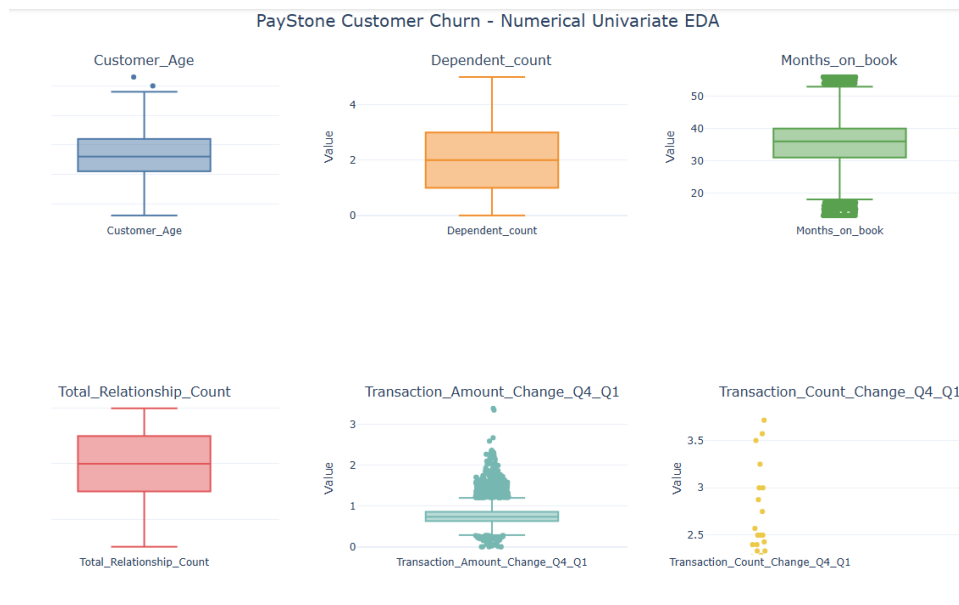


Figure A6. Distribution (boxplots) of core numerical features.

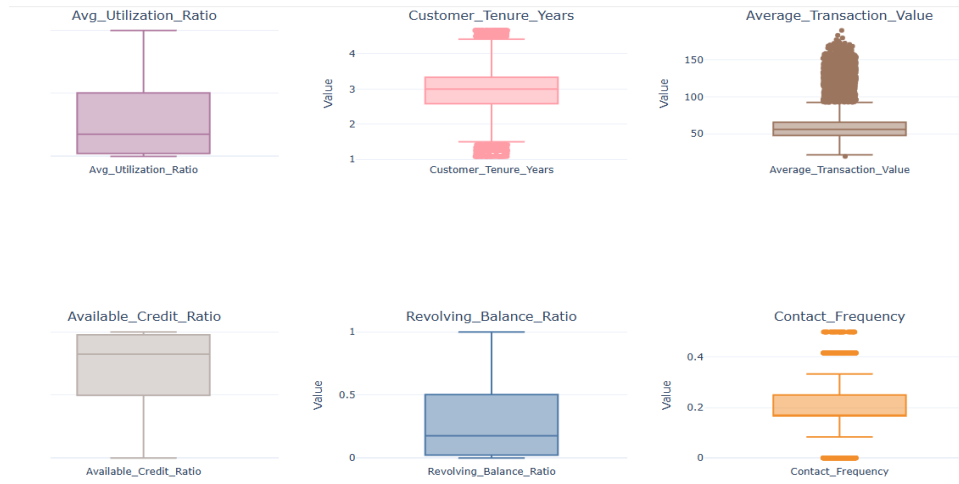


Figure A7. Distribution (boxplots) of engineered ratio features.

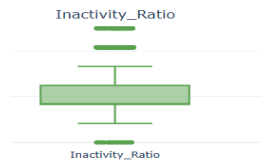


Figure A8. Distribution of Inactivity Ratio.

### Numerical Feature Distribution

- Customer age and dependent count are relatively symmetric, with no significant outliers, indicating stable distributions across the customer base.
- Months on book and customer tenure are concentrated around approximately 36 months (three years), with a small number of lower-end outliers representing newer customers.
- Total relationship count is relatively evenly distributed across its bounded between 1 to 6 range, indicating variation in the number of products held by customers.
- Transaction amount change quarter 4 to quarter 1 and transaction count change quarter4 to quarter 1 are right-skewed and contain outliers, reflecting a smaller group of customers experiencing unusually large changes in transaction activity.
- Average utilisation ratio and revolving balance ratio have virtually identical distributions, reinforcing their perfect correlation and indicating that they provide redundant information.
- Available credit ratio shows the inverse distribution of utilisation, consistent with its mathematical relationship with credit utilisation.

- Average transaction value is right-skewed, with most transactions concentrated at lower values and a long tail of higher-value transactions.
- Contact frequency and inactivity ratio have relatively low variation and display distinct clusters because they take only a limited number of discrete values rather than varying continuously.
- Overall, the numerical features are generally well-behaved and suitable for modelling, with the main preprocessing consideration being the redundancy among the utilisation-related variables and the presence of outliers in transaction-change measures.
- Customer behaviour varies meaningfully across the customer base, with no significant extreme outliers.
- Distinct clusters at the boundaries suggest different engagement levels, supporting targeted customer segmentation and retention strategies.

## A.7 Data Preparation Summary

The modelling dataset comprised 10,127 customer records and 19 pre-expansion variables, prepared through the following steps:

- Removed customer identifiers and non-predictive fields to ensure the model was trained only on relevant customer characteristics.
- Reviewed data completeness and quality, confirming no substantive missing values. The Unknown income category was treated as missing rather than as a genuine income classification.
- Removed duplicate records and reset the dataset index to maintain data integrity and consistency.
- Engineered six behavioural features: tenure in years, average transaction value, available credit ratio, revolving balance ratio, contact frequency, and inactivity ratio. The underlying raw variables were subsequently removed to reduce redundancy and prevent data leakage.
- Encoded categorical variables appropriately: binary encoding for Churn and Gender, one-hot encoding for Education Level and Marital Status, and ordinal encoding for Income Category and Card Category.